Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 4 (27.11.2020)

Problem 4.1: Wave propagation inside metal waveguide

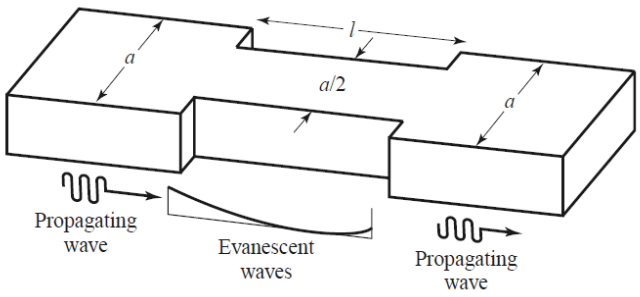
An air-filled rectangular metal waveguide of width 5 cm (x-axis) and height 2 cm (y-axis) has the following field distribution along its longitudinal direction (z-axis) at 15 GHz:

$$E_z = 20\sin(40\pi x)\sin(50\pi y)\, e^{-j\beta z}\ V/m$$

Find
a) the mode that is being propagated
b) the value of β
c) the ratio E_y/E_x as a function of the spatial coordinates (x, y) only.

Problem 4.2: Attenuation in metal waveguides below cutoff

An attenuator can be made using a section of waveguide operating in dominant mode TE_{10} below cutoff, as shown below. If a = 2.286 cm and the operating frequency is 12 GHz, determine the required length of the below-cutoff section of waveguide to achieve an attenuation of 100 dB between the input and output guides. Ignore the effect of reflections at the step discontinuities.



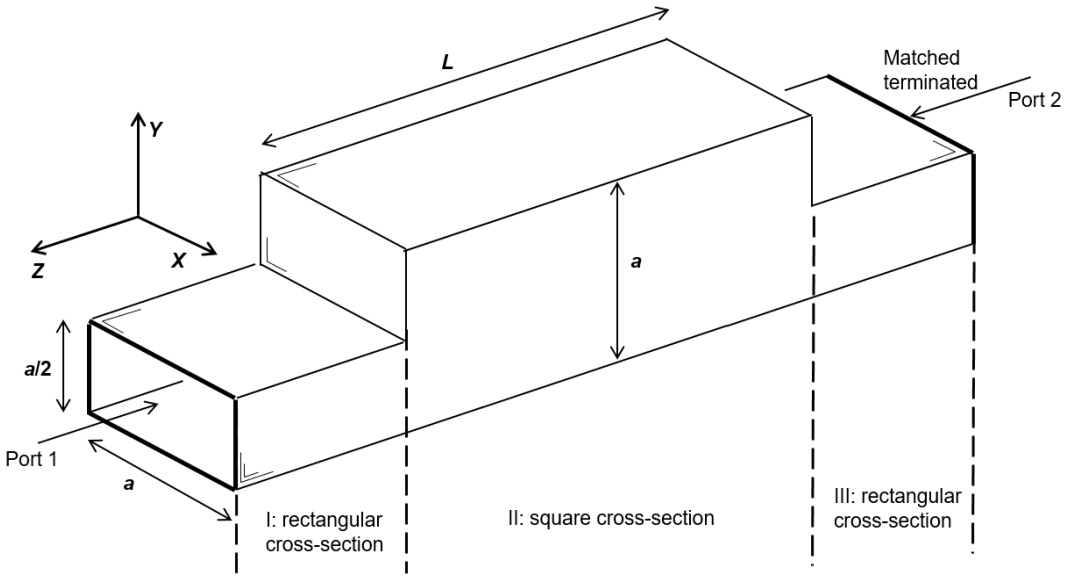
Problem 4.3: Guided wavelength

An air-filled, metallic rectangular waveguide with its width (a) and height (a/2).

a) Calculate the cutoff frequency f_c of the fundamental wave TE_{10} and the maximum frequency for the monomode frequency range.
b) Which mode is limiting the monomode frequency range?

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Problem 4.4: Impedance transformation in metal waveguides



A 2-port air-filled metal waveguide structure shown in the picture above and the circuit operates in TE_{01} mode. Assume there is no conduction loss or dielectric loss in the whole circuit. Port 2 is matched terminated.

a) Calculate the cut-off frequency of the TE_{01} mode in section I and section II of the waveguide circuit.
b) Calculate the wave impedance of the TE_{01} mode in section I and section II of the waveguide circuit at frequency $f = 1.2\ f_{c1}$, where f_{c1} is the cut-off frequency of section I.
c) Draw the transmission line equivalent model of this circuit.
b) Determine length L (smallest possible, but different to zero) for which port 1 is perfectly matched.